Roger Gonzalez Sedano

M.Sc. Telecommunications Engineer — EM Wave Propagation & Modeling

(+34) 646 008 255 — rogonzalezse@gmail.com — Mataró, Spain (Open to Relocate) — profile.rogsbox.com

Telecommunications Engineer specializing in **Electromagnetic Wave Propagation** and **Physical Modeling**. Combines a deep understanding of **RF physics**, gained through research at **KTH** and **N3Cat**, with a strong academic foundation in **Machine Learning**. Passionate about applying computational methods to solve complex scientific challenges and bridging the gap between theoretical research and practical implementation.

TECHNICAL SKILLS

Electromagnetics:	Wave Propagation Modeling, Antenna Theory, CST Studio Suite, Computational EM.
Scientific Computing:	MATLAB (Phased Arrays, Signal Processing), Python (NumPy, SciPy, Matplotlib).
Data Science & AI:	PyTorch, TensorFlow, Statistical Analysis.
Lab & Validation:	VNA (S-Parameters), Spectrum Analyzers, Oscilloscopes, Altium Designer.
Software & Tools:	Git, Docker, LaTeX, PostgreSQL, Embedded Systems (STM32/ESP32).
Network Engineering:	TCP/IP, OpenWRT, OpenWisp, Zabbix, Linux Administration, routing.

WORK EXPERIENCE

Research Intern (M.Sc. Thesis) — **Aug 2024 – May 2025**
KTH Royal Institute of Technology — *Stockholm, Sweden*
- Developed novel mathematical models for **electromagnetic wave propagation** in 6G near-field scenarios.
- Redefined **boundary conditions** and validated theoretical limits of near-field interactions using numerical analysis.
- Implemented extensive simulations in **MATLAB** to bridge the gap between physical channel models and estimation algorithms.
- **Thesis:** Mathematical modeling for near-field 6G communications.

Research Intern — **Sep 2023 – Jun 2024**
NaNoNetworking Center in Catalonia (N3Cat) — *Barcelona, Spain*
- Simulated **EM wave interaction** within complex 3D structures for sub-THz Wireless Network-on-Chip.
- Performed **computational EM simulations** (CST Studio Suite) to optimize radiation efficiency and coupling.
- Investigated Time Reversal techniques to analyze **multipath propagation** and scattering in reverberant environments.

Hardware Engineer Intern — **Apr 2022 – Jul 2022**
Idneo — *Mollet del Vallès, Spain*
- Conducted **laboratory validation** of automotive ECUs using VNAs and Oscilloscopes.
- Performed signal integrity measurements and documented compliance with rigorous technical standards.

Technical Apprentice — **Jul 2021 – Sep 2021**
Engi-on Automatica — *Mollet del Vallès, Spain*
- Developed computer vision software for automated defect detection.
- Authored technical manuals, streamlining maintenance procedures.

EDUCATION

M.Sc. in Advanced Telecommunication Technologies — **Sep 2023 – Jun 2025**
Universitat Politècnica de Catalunya (UPC) — *Barcelona, Spain*
- **Focus:** Wireless Communications, Computational EM, AI for Signal Processing.
- **Relevant Coursework:** Optical Remote Sensing (LIDAR), Advanced Communications for Wireless Systems, Machine Learning from Data, Computer Vision with Deep Learning, 5G Mobile Communications.
- **Project:** Designed ML classifiers for Dynamic Spectrum Access to identify modulation types.

B.Sc. in Telecommunications Technologies — **Sep 2018 – Jan 2023**
Universitat Politècnica de Catalunya (UPC) — *Barcelona, Spain*
- **Relevant Coursework: Microwaves (Honors/MH)**, Space Telecommunications, Radiation & Propagation, Electromagnetic Waves, Antennas, Radio Communications, Optical Communications.
- **Thesis:** *Planar antenna design and channel modeling for chip-scale communications.*

PROJECTS & INTERNATIONAL COOPERATION

4Forms - Service Forms Automation Platform — **Sep 2025 – Dec 2025**
Developed a Django/PostgreSQL platform that replaces paper workflows for field service assistances.
- **Impact:** Reduced administrative processing time by **90%** (from 15 mins to 1 min per form).
- Implemented REST APIs for automated PDF generation and client billing, currently handling daily logs for field technicians.

Namaqua & Hahatay Community Networks (Namibia/Senegal)
Co-leading technical deployment of rural connectivity solutions in collaboration with NGOs.
- **Since 2022:** Designed a network deployed with OpenWRT routers and a centralized control layer (OpenWisp/Zabbix) to provide internet access to **200+ users** in rural Senegal. (hahatay.network).
- **Since 2025:** Replicating the model in Namibia to connect the Namaqua Kalahari Children's Home to essential digital services. (Namaqua project).

CERN Challenge Based Innovation
Selected for a multidisciplinary innovation program at CERN (IdeaSquare). Collaborated with MBA and design students to prototype a disruptive social innovation project on sustainable mobility, leveraging CERN detector technologies.